Project planning phase

1.planning logic:

1. Problem Identification Planning

The first step is to clearly define the problem.
The objective is to predict whether a startup will be successful or not.
This is identified as a classification problem.
The target variable is startup success status.
Proper problem definition helps in selecting the right approach.

2. Data Collection Planning

Relevant startup data is collected in CSV format.
The dataset must contain important business features.
Data should be reliable and sufficient for training.
Both input features and target output must be available.
Good quality data ensures better model accuracy.

3.Data Preprocessing Planning

Raw data needs cleaning before training.
Missing values and duplicate records are handled.
Categorical data is converted into numerical format.
Feature scaling is applied if required.
Clean data improves model performance.

4. Exploratory Data Analysis Planning

EDA is planned to understand patterns in data.
Statistical summaries are calculated.
Graphs and visualizations are used for analysis.
Feature relationships with startup success are studied.
Important influencing factors are identified.

5. Model Selection Planning

Different algorithms are considered for prediction.
Decision Tree, SVM, and Logistic Regression are analyzed.
Random Forest is selected for better accuracy and stability.
The algorithm should reduce overfitting.
The chosen model must handle multiple features effectively.

6. Model Training Planning

The dataset is divided into training and testing sets.
The model is trained using training data.
It learns patterns from historical startup records.
Multiple decision trees are built internally.
Training prepares the model for prediction.

7. **Hyperparameter Tuning Planning**

Model parameters are optimized for better performance.
Different combinations of parameters are tested.
Grid Search or Random Search methods are applied.
Accuracy improvement is monitored carefully.
The best configuration is selected.

8.Model Evaluation Planning

The trained model is tested on unseen data.
Evaluation metrics like accuracy and precision are calculated.
Confusion matrix is analyzed.
Performance before and after tuning is compared.
This ensures the model generalizes well.

9.Model Saving Planning

The best-performing model is finalized.
It is saved as a .pkl file using pickle.
The saved model can be reused anytime.
This avoids retraining the model repeatedly.
It is prepared for deployment.

10.**Deployment Planning**

The model is integrated into a Flask web application.
Routing and templates are designed.
User inputs are collected through web forms.
Input data is processed and passed to the model.
Prediction results are displayed to the user.